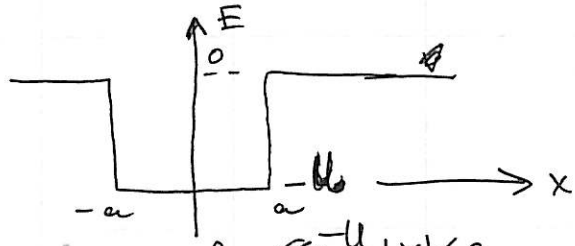
let us do this with <u>even</u> and <u>odd</u> functions



$$V(x) = \begin{cases} -U, & |x|<a \\ 0, & \text{otherwise} \end{cases}$$

Parametrize $E = -\frac{\hbar^2 \varkappa^2}{2m}$ $\quad U = \frac{\hbar^2 k_0^2}{2m}$ $\quad k^2 = k_0^2 - \varkappa^2$

We are looking for <u>bound states</u> (states with $E<0$ energies)

$\Rightarrow \quad E<0 \Rightarrow$ parametrize $E = -\frac{\hbar^2 \varkappa^2}{2m}$

① S.E. for $|x|<a$: $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - U\psi = E\psi$

$$\frac{d^2\psi}{dx^2} = \left(U - \frac{\hbar^2\varkappa^2}{2m}\right)\left(-\frac{2m}{\hbar^2}\right)\psi$$

$$= (\varkappa^2 - k_0^2)\psi = -(k_0^2 - \varkappa^2)\psi$$

denote $k^2 = k_0^2 - \varkappa^2 > 0 \Rightarrow U+E>0$, $E>-U$ <u>ok</u>

$$\Rightarrow \frac{d^2\psi}{dx^2} = -k^2\psi \qquad |x|<a$$

② Outside $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = -\frac{\hbar^2\varkappa^2}{2m}\psi$ (because $V(x)=0$)

$$\Rightarrow \frac{d^2\psi}{dx^2} = \varkappa^2\psi \qquad |x|>a$$

Divide solutions into <u>even</u> and <u>odd</u>

<u>even</u> : $\psi(x) = \psi(-x)$

$$\Rightarrow \psi(x) = \begin{cases} A\cos kx & |x|<a \\ A\cos ka\, e^{\varkappa(a-x)}, & x>a \\ A\cos ka\, e^{\varkappa(a+x)}, & x<a \end{cases}$$

I used continuity of the w.f. ← here

Clearly $\psi(a-0) = A\cos ka = \psi(a+0) = A\cos ka$

<u>odd</u>

$$\psi(x) = \begin{cases} B\sin kx, & 0<x<a \\ B\sin ka\, e^{\varkappa(a-x)}, & x>a \end{cases}$$

$$\psi(-x) = -\psi(x)$$

A and B can be found from $\int_{-\infty}^{\infty} dx\, |\psi(x)|^2 = 1$.

So constructed, w.f. $\psi(x)$ is continuous everywhere

$\Rightarrow$ ~~Now~~ use continuity of $\psi'(x)$ as well:

$x=a$ $\quad \underline{\psi'(a-0) = \psi'(a+0)}$

<u>EVEN</u> $\psi' = \begin{cases} -Ak\sin kx \\ -A\varkappa\cos ka\, e^{\varkappa(a-x)} \end{cases}$

$$\Rightarrow \quad -Ak\sin ka = -A\varkappa\cos ka$$

$$\boxed{\tan(ka) = \frac{\varkappa}{k}} \qquad (1)$$

<u>ODD</u> $\psi' = \begin{cases} Bk\cos kx \\ -B\varkappa\sin ka\, e^{\varkappa(a-x)} \end{cases}$

$$\Rightarrow \quad Bk\cos ka = -B\varkappa\sin ka$$

$$\boxed{\cot(ka) = -\frac{\varkappa}{k}} \qquad (2) \quad \Rightarrow \quad \tan(ka) = -\frac{k}{\varkappa}$$

<u>SOLVE (1)</u> $\quad \tan(ka) = \frac{\varkappa a}{ka}$

<u>EVEN</u>

$$\varkappa a = \sqrt{(k_0 a)^2 - (ka)^2} = \sqrt{C^2 - X^2}$$

$$ka \equiv X, \quad k_0 a \equiv C$$

$$\Rightarrow \quad \tan(X) = \frac{\sqrt{C^2 - X^2}}{X} \quad \text{where } C = k_0 a = \sqrt{\frac{2mU}{\hbar^2}}\, a$$

<u>ODD</u> $\quad \cancel{\tan}(x) = -\dfrac{x}{\sqrt{c^2-x^2}} \quad (2')$

<u>EVEN</u> $\quad \tan(x) = \dfrac{\sqrt{c^2-x^2}}{x} \quad (1')$

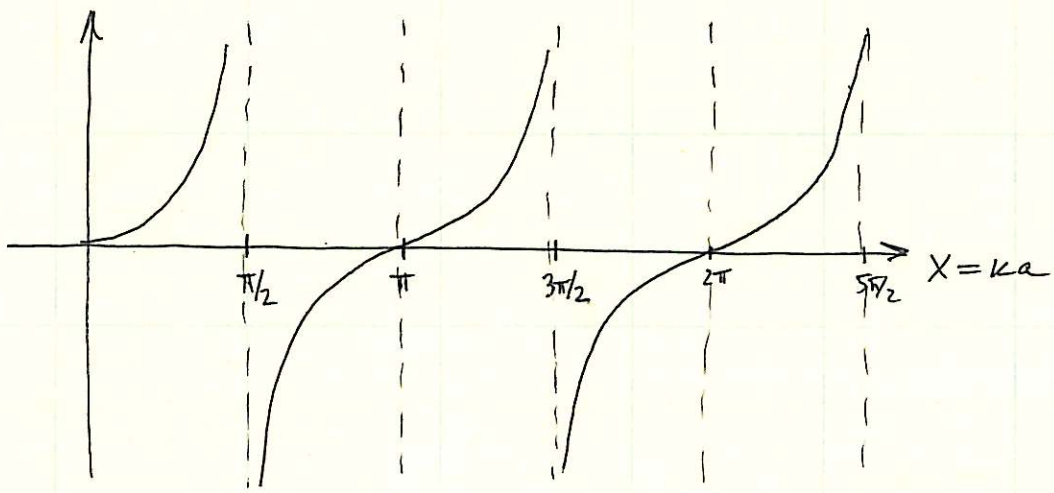
→ Once $\underline{\underline{x}}$ solving (1') and (2') are found

$$E = -\frac{\hbar^2}{2ma^2}(\varkappa a)^2 = -\frac{\hbar^2}{2ma^2}\left(c^2-x^2\right) = -\frac{\hbar^2}{2ma^2}\left(\frac{2mUa^2}{\hbar^2} - x^2\right)$$

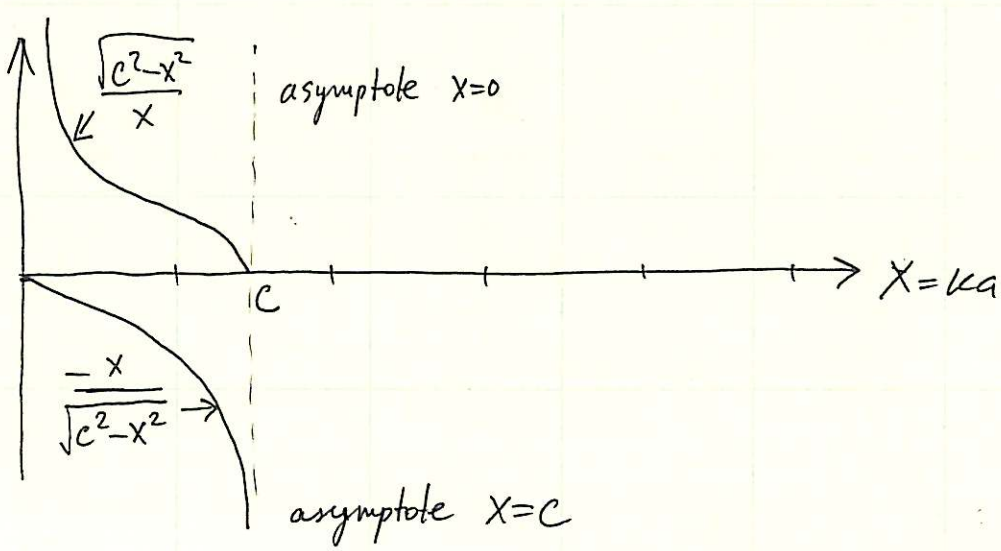
$$= -U\left[1-\left(\frac{x}{c}\right)^2\right] \qquad c^2 = \frac{2mUa^2}{\hbar^2}$$

$$E > -U \Rightarrow -U\left[1-\left(\frac{x}{c}\right)^2\right] > -U$$

$$1-\left(\frac{x}{c}\right)^2 < 1 \;;\; \left(\frac{x}{c}\right)^2 > 0 \quad \underline{\underline{\text{ok}}}$$



$2+\frac{1}{2} = \frac{5\pi}{2}$



It is easy to see the following facts:

(1) for $0 < C < \frac{\pi}{2}$ there is only 1 solution in the <u>EVEN</u> sector (and none in the ODD).

(2) for $0 < C < \pi$: 1 solution in EVEN
1 — " — in ODD

(3) $\pi < C < \frac{3\pi}{2}$: 2 solutions in EVEN
1 in ODD

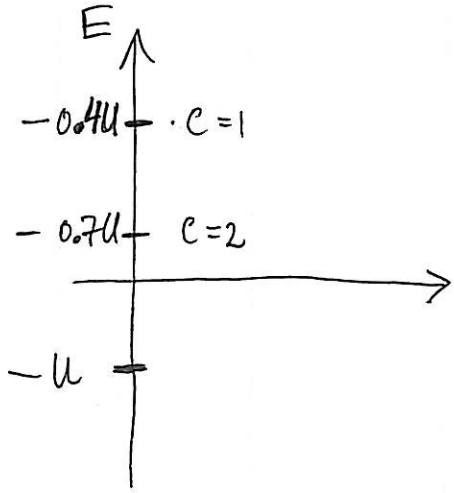
Thus, for $0 < C < \frac{5\pi}{2}$: 3 solutions in <u>EVEN</u>
2 in <u>ODD</u>

E.g. for $C = 1$ (this is $< \pi/2 \simeq 1.57$) $x_1 \cong 0.75$

$$\Rightarrow E_1 = -U\left[1 - (0.75)^2\right] = -U\left[0.44\right]$$

for $C = 2$ $x_1 \simeq 1.03$ (even)

$$E_1 = -U\left[1 - \left(\frac{1.03}{2}\right)^2\right] = -U\left[0.74\right]$$



$x_2 \simeq 1.9$ (odd)

$$E_2 = -U\left[1 - \left(\frac{1.9}{2}\right)^2\right] = -U\left[0.08\right]$$

Also NOTE that the lowest energy that the ground state can have corresponds to $x_1 = \pi/2$

when $\tan(x_1) = \frac{\sqrt{C^2 - x_1^2}}{x_1} = \infty$

$$\Rightarrow C = \infty \Rightarrow \cos x = 0$$

$$\Rightarrow x_1 = k_1 a = \frac{\pi}{2} \quad \Rightarrow \quad (\varkappa a)^2 = (k_0 a)^2 - (k a)^2 =$$

$$= (k_0 a)^2 - \left(\frac{\pi}{2}\right)^2$$

The corresponding energy is

$$E_1 = -\frac{\hbar^2}{2ma^2}(\varkappa a)^2 = -\frac{\hbar^2 k_0^2 a^2}{2ma^2} + \frac{\hbar^2}{2ma^2}\cdot\frac{\pi^2}{4}$$

$$= \frac{\hbar^2\pi^2}{8ma^2} - U = \frac{\hbar^2\pi^2}{2m(2a)^2} - U$$

This is the ground state of the $V_0 = \infty$ well with $L = 2a$ $\left(E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}\right)$

As $0 < x_1 < \frac{\pi}{2}$ and

$E_1 = -\frac{\hbar^2}{2ma^2}x_1^2$, the level appears with $E_1 = -0$ and moves down as the depth of the well $U = \frac{\hbar^2}{2m}\left(\frac{c}{a}\right)^2$ is increased.

For the second level to appear (this happens in the <u>ODD</u> sector)

$$c \geq \frac{\pi}{2} \Rightarrow U \geq \frac{\hbar^2}{2m}\left(\frac{\pi}{2a}\right)^2 = \frac{\hbar^2\pi^2}{8ma^2} \quad (\text{which is } E_2!)$$

Note that $E = -U\left(1 - \frac{x^2}{c^2}\right) = -U + U\frac{x^2}{c^2} = \left(c^2 = \frac{2ma^2}{\hbar^2}U\right) =$

$$= -U + \frac{\hbar^2 x^2}{2ma^2}$$

$$\boxed{E = -U + \frac{\hbar^2 x^2}{2ma^2}}$$

Suppose small $x \ll \frac{\pi}{2} \Rightarrow \tan(x) \approx x = \frac{\sqrt{c^2 - x^2}}{x}$

$$\Rightarrow x^4 = c^2 - x^2$$

$$u = x^2: \quad u^2 + u - c^2 = 0 \qquad u_{1,2} = \frac{-1 \pm \sqrt{1+4c^2}}{2}$$

take $u_1(+) > 0$ only

$$\Rightarrow x^2 = \frac{\sqrt{1+4c^2} - 1}{2} \approx c^2 - c^4 \quad \left(\text{for } C \to 0, \text{ i.e. } u \to 0\right)$$

$$E + U = U \frac{x^2}{c^2} = \cancel{\frac{U}{\frac{2mUa^2}{\hbar^2}} x^2 = U \frac{\hbar^2}{2ma^2}}$$

$$= U \frac{c^2 - c^4}{c^2} = U\left(1 - c^2\right)$$

$$\Rightarrow E = -Uc^2 = -\frac{2ma^2}{\hbar^2} U^2 = -\frac{m}{2\hbar^2}\left(U \cdot 2a\right)^2$$

$$= -\frac{m}{2\hbar^2}\left(\int_{-\infty}^{\infty} dx\, U(x)\right)^2 \leftarrow \text{eq. after (1) p. 201 of Landau, Lifshitz}$$

because $\int_{-\infty}^{\infty} dx\, U(x) = -U \cdot 2a$

$\Rightarrow$ the energy level $\cancel{E_1 = -\frac{m}{2\hbar^2}(2aU)^2}$

$$|E_1| = +\frac{2ma^2}{\hbar^2} U^2 \ll U$$

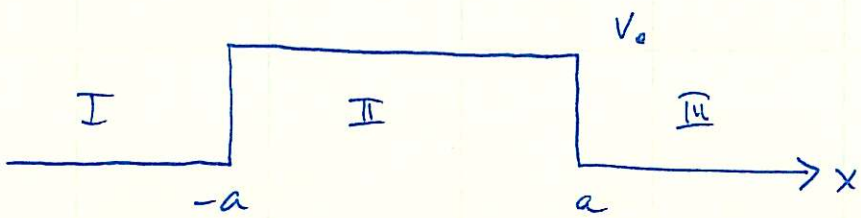
is more shallow than the well!

But is <u>present for any</u> U! $\leftarrow$ this is special feature of $D = 1, 2$.

<u>Units</u>: $[E_1] = \frac{kg \cdot m^2}{(J \cdot s)^2} J^2 = \frac{kg \cdot m^2}{s^2} =$

$$= \frac{kg \cdot m}{s^2} \cdot m = N \cdot m = J$$

Note that $\frac{ma^2}{\hbar^2} U$ = dimensionless.



$$V(x) = \begin{cases} V_0 & |x|<a \\ 0 & |x|>a \end{cases}$$

① $0 < E < V_0$: tunneling

② $E = V_0$

③ $E > V_0$: scattering states everywhere.

↳ obtained by $V_0 \to -V_0$ replacement! in eq. [2.169]

$$T = \left\{ 1 + \frac{V_0^2}{4E(E-V_0)} \sin^2 \frac{2a}{\hbar}\sqrt{2m(E-V_0)} \right\}^{-1}$$

⇒ again: perfect transmission when

$$E_s - V_0 = \frac{\pi^2 s^2 \hbar^2}{8ma^2}$$

① $$\psi = \begin{cases} A e^{ikx} + B e^{-ikx} & \text{I} \\ C e^{\varkappa x} + D e^{-\varkappa x} & \text{II} \\ F e^{ikx} & \text{III} \end{cases} \qquad k = \sqrt{\frac{2mE}{\hbar^2}} \qquad \varkappa = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$$

$X=-a$

$$A e^{-ika} + B e^{ika} = C e^{-\varkappa a} + D e^{\varkappa a}$$

$$ik\left(A e^{-ika} - B e^{ika}\right) = \varkappa\left(C e^{-\varkappa a} - D e^{\varkappa a}\right)$$

$$\Rightarrow 2A e^{-ika} = C e^{-\varkappa a}\left(1 - \frac{i\varkappa}{k}\right) + D e^{\varkappa a}\left(1 + \frac{i\varkappa}{k}\right)$$

$X=a$

$$\cancel{A e^{ika} + B e^{-ika} = C e^{\varkappa a} + D e^{-\varkappa a}}$$

$$\cancel{A e^{ika}}$$

$$F e^{ika} = C e^{\varkappa a} + D e^{-\varkappa a}$$

$$ik F e^{ika} = \varkappa\left(C e^{\varkappa a} - D e^{-\varkappa a}\right)$$

$$\Rightarrow \begin{cases} 2C e^{\varkappa a} = F e^{ika}\left(1 + \frac{ik}{\varkappa}\right) \\ 2D e^{-\varkappa a} = F e^{ika}\left(1 - \frac{ik}{\varkappa}\right) \end{cases}$$

$$\Rightarrow 2A e^{-ika} = \frac{1}{2} F e^{ika} e^{-2\varkappa a}\left(1 + \frac{ik}{\varkappa}\right)\left(1 - \frac{i\varkappa}{k}\right)$$

$$+ \frac{1}{2} F e^{ika} e^{2\varkappa a}\left(1 - \frac{ik}{\varkappa}\right)\left(1 + \frac{i\varkappa}{k}\right) =$$

$$= \frac{1}{2} F e^{ika}\left\{e^{-2\varkappa a}\left(1 + 1 - i\left(\frac{\varkappa}{k} - \frac{k}{\varkappa}\right)\right)\right.$$

$$\left. + e^{2\varkappa a}\left(2 + i\left(\frac{\varkappa}{k} - \frac{k}{\varkappa}\right)\right)\right\} =$$

$$= \frac{1}{2} F e^{ika}\left\{4\,ch(2\varkappa a) + i\,\frac{\varkappa^2 - k^2}{k\varkappa}\,2\,sh(2\varkappa a)\right\}$$

$$= 2F e^{ika}\left\{ch(2\varkappa a) + i\,\frac{\varkappa^2 - k^2}{2k\varkappa}\,sh(2\varkappa a)\right\}$$

$$T^{-1} = \left|\frac{A}{F}\right|^2 = \left| e^{i2ka}\left(ch(2\varkappa a) + i\,\frac{\varkappa^2-k^2}{2k\varkappa}\,sh(2\varkappa a)\right)\right|^2$$

$$= ch^2(2\varkappa a) + \left(\frac{\varkappa^2-k^2}{2k\varkappa}\right)^2 sh^2(2\varkappa a)$$

$$ch^2 x = 1 + sh^2 x$$

$$T^{-1} = 1 + \left[1 + \left(\frac{\varkappa^2-k^2}{2k\varkappa}\right)^2\right] sh^2(2\varkappa a) =$$

$$= 1 + \left(\frac{\varkappa^2+k^2}{2k\varkappa}\right)^2 sh^2(2\varkappa a)$$

$$= 1 + \left(\frac{\frac{2m}{\hbar^2}(V_0 - E + E)}{2\,\frac{2m}{\hbar^2}\sqrt{E(V_0-E)}}\right)^2 sh^2\,\frac{2a}{\hbar}\sqrt{2m(V_0-E)}$$

$$= 1 + \frac{V_0^2}{4E(V_0-E)}\,sh^2\left(\frac{2a}{\hbar}\sqrt{2m(V_0-E)}\right).$$

$$T = \left(1 + \frac{V_0^2}{4E(V_0-E)}\,sh^2\,\frac{2a}{\hbar}\sqrt{2m(V_0-E)}\right)^{-1} \leftarrow \text{answer}$$

let $\underline{V_0 - E = \text{fixed} > 0}$ and increase the width $\underline{a}$

$$sh(at) = \frac{1}{2}\left(e^{at} + e^{-at}\right) \to \frac{1}{2}e^{at}$$

$$T \underset{a\to\infty}{\to} \left(1 + \frac{V_0^2}{16E(V_0-E)}\,e^{\frac{2a}{\hbar}\sqrt{2m(V_0-E)}}\right)^{-1} =$$

$$= \frac{16E(V_0-E)}{V_0^2} e^{-\frac{2a}{\hbar}\sqrt{2m(V_0-E)}} \xrightarrow[a\to\infty]{} 0$$

Transmission decays exponentially with the width a.

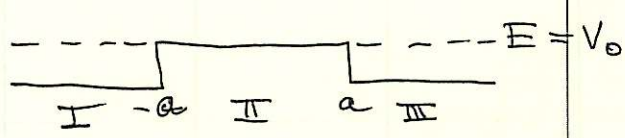
Also recall that $\varkappa = \frac{\sqrt{2m(V_0-E)}}{\hbar} = \frac{\text{momentum}}{\hbar}$

$$\Rightarrow T \sim \exp\left(-\frac{1}{\hbar}\int dx\, p(x)\right)$$

quasi-classical approximation

the integral is over classically forbidden region.

(2) let $E = V_0$ $\Rightarrow$ we have scattering states in regions I and III



$$\psi = \begin{cases} Ae^{ikx} + Be^{-ikx}, & \text{I} \\ C + Dx, & \text{II} \\ Fe^{ikx}, & \text{III} \end{cases} \qquad k = \sqrt{\frac{2mE}{\hbar^2}} = \sqrt{\frac{2mV_0}{\hbar^2}}$$

Inside the barrier $E = V_0$ the Schrodinger equation reads

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V_0\psi = V_0\psi \quad (\text{for } x \in \text{II})$$

$$\Rightarrow \frac{d^2\psi}{dx^2} = 0 \Rightarrow \psi = C + Dx$$

Boundary conditions: $\underline{x=-a}$ $\quad Ae^{-ika} + Be^{ika} = C - Da$

$$ik\left(Ae^{-ika} - Be^{ika}\right) = D$$

$$\Rightarrow 2Ae^{-ika} = C - Da + \frac{D}{ik} = C - D\left(a + \frac{i}{k}\right)$$

<u>At $x=a$</u>

$$\left.\begin{aligned} C + Da &= Fe^{ika} \\ D &= ikFe^{ika} \end{aligned}\right\} \quad \begin{aligned} C &= Fe^{ika} - ikaFe^{ika} \\ &= Fe^{ika}(1-ika) \end{aligned}$$

Thus from $2Ae^{-ika} = C - D\left(a + \frac{i}{k}\right)$

$$\Rightarrow \quad 2Ae^{-ika} = Fe^{ika}(1-ika) - ikFe^{ika}\left(a+\frac{i}{k}\right) =$$

$$= Fe^{ika}\left(1 - ika - ika + 1\right)$$

$$= 2Fe^{ika}(1-ika)$$

$$T^{-1} = \left|\frac{A}{F}\right|^2 = \left|e^{i2ka}(1-ika)\right|^2 = 1 + k^2a^2 = 1 + \frac{2mV_0a^2}{\hbar^2}$$

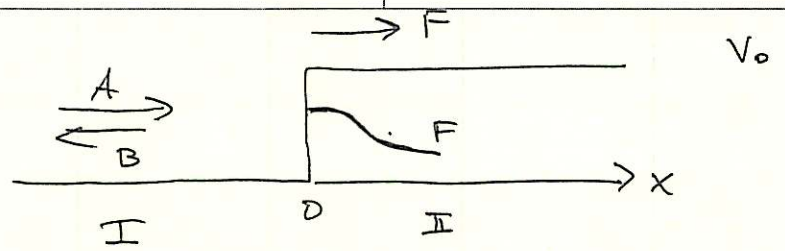
transmission $\quad T = \dfrac{1}{1 + \dfrac{2mV_0a^2}{\hbar^2}}$

Thus

$$T = \left\{1 + \frac{V_0^2}{4E(E-V_0)} \sin^2 \frac{2a}{\hbar}\sqrt{2m(E-V_0)}\right\}^{-1} \quad \text{for } E > V_0 \text{ (scattering)}$$

$$T = \left\{1 + \frac{2mV_0a^2}{\hbar^2}\right\}^{-1} \quad \text{for } E = V_0$$

$$T = \left\{1 + \frac{V_0^2}{4E(V_0-E)} \sinh^2 \frac{2a}{\hbar}\sqrt{2m(V_0-E)}\right\}^{-1} \quad \text{for } 0 < E < V_0 \text{ (tunneling)}$$



$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

$$\Rightarrow \begin{cases} \dfrac{d^2\psi}{dx^2} = \dfrac{2m}{\hbar^2}(V_0 - E)\psi & x \in \mathrm{II} \\[2ex] \dfrac{d^2\psi}{dx^2} = -\dfrac{2mE}{\hbar^2}\psi & x \in \mathrm{I} \end{cases}$$

(a) $0 < E < V_0 \Rightarrow$ denote : $\varkappa = \sqrt{\dfrac{2m}{\hbar^2}(V_0 - E)}$ $\quad k = \sqrt{\dfrac{2mE}{\hbar^2}}$

$\psi_{\mathrm{I}} = A e^{ikx} + B e^{-ikx}$ $\quad$ ($x < 0$, region I)

$\psi_{\mathrm{II}} = F e^{-\varkappa x}$ $\quad$ ($x > 0$, region II) exponential decay under barrier

BC
$$\begin{cases} A + B = F = -\dfrac{ik}{\varkappa}(A - B) \\ ik(A - B) = -\varkappa F \end{cases}$$

$$\Rightarrow \quad A\left(1 + \frac{ik}{\varkappa}\right) = -B\left(1 - \frac{ik}{\varkappa}\right)$$

Reflection coefficient $\quad R = \left|\dfrac{B}{A}\right|^2 = \left|\dfrac{1 + \frac{ik}{\varkappa}}{-\left(1 - \frac{ik}{\varkappa}\right)}\right|^2 = \dfrac{1 + \frac{k^2}{\varkappa^2}}{1 + \frac{k^2}{\varkappa^2}} = 1$

$\Rightarrow$ full reflection, as expected, because for $E < V_0$ only exponentially decaying w.f. exist for $x > 0$.

(b) $E > V_0 \Rightarrow$ denote $l = \sqrt{\dfrac{2m}{\hbar^2}(E - V_0)}$

$$\begin{cases} \psi_{\mathrm{I}} = A e^{ikx} + B e^{-ikx} & x < 0 \\ \psi_{\mathrm{II}} = F e^{ilx} & x > 0 \end{cases}$$

BC
$$\begin{cases} A+B = F = \frac{ik}{il}(A-B) \Rightarrow A\left(1-\frac{k}{l}\right) = -B\left(1+\frac{k}{l}\right) \\ ik(A-B) = ilF \end{cases}$$

$$R = \left|\frac{B}{A}\right|^2 = \left|\frac{1-\frac{k}{l}}{1+\frac{k}{l}}\right|^2 = \left(\frac{l-k}{l+k}\right)^2 = \frac{(k-l)^2(k-l)^2}{(k+l)^2(k-l)^2} =$$

$$= \frac{(k-l)^4}{(k^2-l^2)^2}$$

$$k-l = \frac{\sqrt{2m}}{\hbar}\left(\sqrt{E}-\sqrt{E-V_0}\right)$$

$$k^2-l^2 = \frac{2m}{\hbar^2}\left(E-(E-V_0)\right) = \frac{2mV_0}{\hbar^2}$$

$$\Rightarrow R = \frac{\left(\sqrt{E}-\sqrt{E-V_0}\right)^4}{V_0^2}$$

(c) Transmission coefficient is defined by

$$T = \frac{J_{trans}}{J_{incom}} \quad \text{where } J_{trans/incom} = \text{transmitted/incoming probability current}$$

$$J_{incom} = \frac{\hbar k}{m}|A|^2 \qquad J_{trans} = \frac{\hbar l}{m}|F|^2$$

Note that $\frac{\hbar k}{m} = \frac{p}{m}$ = "velocity" for $x<0$ region

$\frac{\hbar l}{m}$ = "velocity" for $x>0$ region

$$\Rightarrow T = \frac{l}{k}\left|\frac{F}{A}\right|^2 = \sqrt{\frac{E-V_0}{E}}\left|\frac{F}{A}\right|^2$$

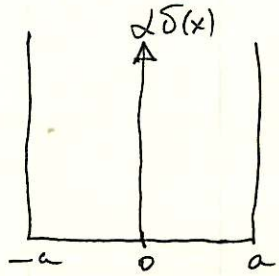
(d) $E>V_0$; find $\frac{F}{A}$: $F = A+B = A\,A\,\frac{1-\frac{k}{l}}{1+\frac{k}{l}} = A\,\frac{1+\frac{k}{l}-1+\frac{k}{l}}{1+\frac{k}{l}} =$

$$= \frac{2k}{k+l}A.$$

$$T = \frac{l}{k}\left|\frac{2k}{k+l}\right|^2 = \frac{4kl}{(k+l)^2} = \frac{4kl(k-l)^2}{(k^2-l^2)^2}$$

$$\Rightarrow T+R = \frac{(k-l)^2}{(k^2-l^2)^2}\left(4kl+(k-l)^2\right) = \frac{(k-l)^2(k+l)^2}{(k^2-l^2)^2} = 1.$$

$$V(x) = \begin{cases} \alpha\,\delta(x) & |x| < a \\ \infty & |x| > a \end{cases} \qquad k = \sqrt{\frac{2mE}{\hbar^2}}$$



Boundary conditions: $\psi(-a) = \psi(a) = 0$

but, integrating around $x=0$, the derivative jumps $\left.\frac{d\psi}{dx}\right|_{+0} - \left.\frac{d\psi}{dx}\right|_{-0} = \frac{2m\alpha}{\hbar^2}\psi(0)$ (2.125) with $\alpha \to -\alpha$

and $\psi(+0) = \psi(-0)$.

(1) Even solutions $\psi(-x) = \psi(x)$.

Thus, if $\psi(x) = A\cos kx + B\sin kx$ for $a > x > 0$

then for $-a < x < 0$ $\psi(x) = \psi(-x) = A\cos k(-x) + B\sin k(-x) =$

$= A\cos kx - B\sin kx$; $-a < x < 0$

$$\psi(+0) = A = \psi(-0) = A \quad \text{(nothing new)}$$

$$\left.\frac{d\psi}{dx}\right|_{+0} = \left(-Ak\sin kx + Bk\cos kx\right)_{x=+0} = Bk$$

$$\left.\frac{d\psi}{dx}\right|_{x=-0} = \left(-Ak\sin kx - Bk\cos kx\right)_{x=-0} = -Bk$$

$$\Rightarrow \quad 2Bk = \frac{2m\alpha}{\hbar^2}A \quad \Rightarrow A = \frac{\hbar^2 k}{m\alpha}B$$

Since $\psi(x) = \psi(-x) \Rightarrow$ if $\psi(a) = 0$ then $\psi(-a) = 0$

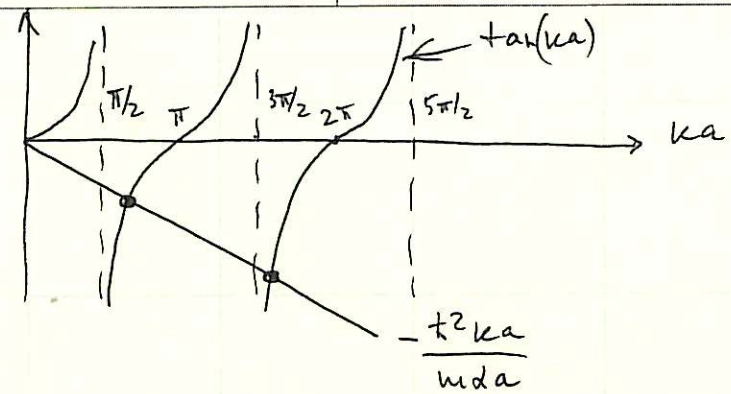
$$\Rightarrow \quad A\cos ka + B\sin ka = 0$$

$$B\left(\frac{\hbar^2 k}{m\alpha}\cos ka + \sin ka\right) = 0$$

$$\Rightarrow \quad \frac{\sin ka}{\cos ka} = -\frac{\hbar^2 k}{m\alpha} = -\frac{\hbar^2 ka}{m\alpha a}$$

this is quantization condition on $k = \frac{\sqrt{2mE}}{\hbar}$

$\psi_{even}(x) = A\cos(kx) + B\sin|kx|$ absolute value!



From the graph, solutions are above (to the right) $\frac{\pi(2n+1)}{2} = \frac{\pi n}{2}$, $n=1,3,5\ldots$ lines.

$$ka > \frac{\pi n}{2} \Rightarrow \frac{2mE_n}{\hbar^2} > \frac{\pi^2 n^2}{4a^2}$$

$$\underline{\underline{\text{even}}} \qquad E_n > \frac{\pi^2\hbar^2 n^2}{8ma^2}$$

i.e. the energies for <u>even</u> solutions are above the corresponding energies of infinite square well $E_n^{well} = \frac{\pi^2\hbar^2 n^2}{8ma^2}$

<u>For $\alpha \to 0$</u> (<u>weak</u> delta-function)

the slope of $-\frac{\hbar^2 ka}{m\alpha a}$ approaches $(-\infty)$

and the intersection points approach $ka \Rightarrow \frac{\pi n}{2}$ (dashed lines)

So that energies approach $E_n^{well} = \frac{\pi^2\hbar^2 n^2}{8ma^2}$ $\left(\text{eq. 2.27 with } a \to 2a\right)$ as they should.

<u>For $\alpha \to \infty$</u> (strong delta-function)

the slope $-\frac{\hbar^2}{m\alpha a} \to 0$ so that intersection points are at $ka = \pi n$!

$$\Rightarrow E_n \to \frac{\pi^2\hbar^2 n^2}{2ma^2}$$

That is, δ-function splits the well into <u>two</u>, each of width <u>a</u>, $\Rightarrow$ hence the energies are given by eq.(2.27) : $E^{well}_{width\,a} = \frac{\pi^2\hbar^2 n^2}{2ma^2}$ $n = 1,3,5\ldots$

(2) <u>Odd</u> solutions $\psi(-x) = -\psi(x)$

For $a > x > 0$: $\psi(x) = A \cos kx + B \sin kx$

$-a < x < 0$: $\psi(x) = -\psi(-x) = (-1)\left[A \cos(-kx) + B \sin(-kx)\right]$

$= -A \cos kx + B \sin kx$

Thus, $\psi(+0) = A = \psi(-0) = -A \Rightarrow \underline{A = 0}$

$$\left.\frac{d\psi}{dx}\right|_{+0} = Bk \cos kx \Big|_{x=+0} = Bk$$

$$\left.\frac{d\psi}{dx}\right|_{x=-0} = Bk \cos kx \Big|_{x=-0} = Bk$$

$$\Rightarrow Bk - Bk = \frac{2m\alpha}{\hbar^2}(0) \quad \text{(nothing new)}$$

$$\psi_{odd}(x) = \begin{cases} B \sin kx & \text{for } a > x > 0 \\ B \sin kx & \text{for } -a < x < 0 \end{cases}$$

Finally, $\psi(+a) = 0 \Rightarrow \psi(-a) = 0$ as well.

$$\psi(+a) = B \sin(ka) = 0 \Rightarrow ka = \pi n \quad n =$$

$$\Rightarrow E_n = \frac{\pi^2 \hbar^2 n^2}{2ma^2}$$

<u>the lesson</u> $\psi_{odd}(0) = 0 \Rightarrow$ delta-function $\delta(x)$ has no effect on odd wavefunctions!